Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 1
Introduction to Blockchain Technology
Estimated Reading Time: 20–25 minutes

Why This Topic Matters

Blockchain technology has emerged as one of the most transformative innovations of the digital era. Initially developed to support cryptocurrencies such as Bitcoin, it is now used in finance, healthcare, supply chain management, digital identity, and many other domains. Understanding its foundations helps learners appreciate how trust can be established without relying on a central authority.

During the first lecture, we will focus on the motivation behind blockchain technology, its evolution, and its core principles rather than implementation details. This pre-reading prepares you for active participation during the classroom discussion.

Learning Outcomes

- Define blockchain in your own words.
- Explain why blockchain technology was developed.
- Differentiate centralized and decentralized systems.
- Identify key characteristics and applications of blockchain.
- Recognize current challenges of blockchain technology.

Activate Your Prior Knowledge

1. Why do we trust banks to store our money?
Think about the role of banks as trusted intermediaries and how transactions are verified.

2. What happens when you make a UPI payment?
Consider how banks validate and record every transaction before completion.

3. Can two strangers exchange money without trusting each other?
Reflect on how trust might be established without a central authority.

4. Why is data tampering a serious problem?
Consider examples such as land records, medical records, or financial transactions.

Prerequisite Concepts

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Basic knowledge of computer networks, databases, and the Internet is sufficient for this lecture. No prior knowledge of blockchain or cryptography is expected.

Core Concepts

What is Blockchain?

Blockchain is a decentralized digital ledger that records transactions across multiple computers. Every participant maintains a synchronized copy of the ledger, making records transparent and resistant to unauthorized modification.

Why Was Blockchain Developed?

Traditional systems rely on trusted intermediaries. Blockchain reduces dependence on such intermediaries by enabling participants to collectively verify transactions.

Centralized vs Decentralized Systems

Centralized systems are controlled by a single organization, whereas decentralized systems distribute control among multiple participants, improving resilience and transparency.

Characteristics of Blockchain

Key characteristics include decentralization, transparency, immutability, security, and traceability.

Applications of Blockchain

Blockchain is applied in cryptocurrencies, banking, healthcare, supply chains, digital identity, education, insurance, and voting systems.

Challenges of Blockchain

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Major challenges include scalability, energy consumption, interoperability, and regulatory uncertainty.

Guided Reading Questions

1. Why was blockchain introduced?
2. Why is decentralization important?
3. Why is blockchain considered immutable?
4. Name applications beyond cryptocurrency.
5. What are the major limitations of blockchain?

Reflection Activity

Write a 150–200 word reflection describing one real-world application where blockchain could improve an existing system. Explain the problem, how blockchain could help, and the expected benefits.

Self-Assessment

1. Define blockchain.
2. Why was blockchain developed?
3. List three characteristics.
4. Differentiate centralized and decentralized systems.
5. Name three blockchain applications.

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